

RE100 Gap Analysis Methodology v0.1

> **PRE-KICKOFF DRAFT / INTERNAL WORKING METHOD** > 本文件不包含已驗證 AMAT-specific consumption、target、site 或 procurement facts。Gap_Analysis_Model.xlsx 中所有數字均為 SYNTHETIC / DEMO DATA。正式專案須以 AMAT policy owner、data owners 與 source evidence 更新。

1. Management intent

本分析不是為了產生一個看似精確的缺口數字，而是建立一個 AMAT 能核准、往後可持續更新的 renewable electricity requirement baseline。管理者應能從成果回答：

1. 分母包含哪些 entities / sites / meters / periods ?
2. target 是哪個 policy version、哪條 curve、由誰核准？
3. 哪些既有採購量被視為 eligible，依據是什麼？
4. residual gap 是單一數字，還是受資料 / delivery 不確定性影響的 range？
5. 哪些 decisions 能進入 1.5，哪些必須先補 evidence？

2. Calculation boundary before calculation

2.1 Boundary card

正式 run 前先完成並核准一頁 boundary card：

Boundary item	Required decision	v0.1 status
Organizational	legal entities / operations included	TBD WITH CLIENT
Geographic	Taiwan only、specific sites 或其他	TBD WITH CLIENT
Load	gross / net、onsite self-use、excluded loads	TBD WITH CLIENT
Time	base year、planning horizon、calendar / fiscal year	TBD WITH CLIENT
Target	percentage curve、target date、policy version	TBD WITH CLIENT
Eligibility	instruments、vintage、ownership、claim rules	TBD WITH CLIENT
Existing procurement	contracted / expected / delivered / eligible	TBD WITH CLIENT
Granularity	annual first pass、monthly / interval if required	HYPOTHESIS

在 boundary 未核准前，model output 只能標為 draft / scenario，不能標為 AMAT baseline。

2.2 Numerator / denominator discipline

- Denominator : approved electricity consumption / forecast after explicit inclusion / exclusion rules °
- Target renewable requirement : denominator × target percentage °
- Numerator / coverage : 只計入符合 approved eligibility rule 且可歸屬於相同 boundary / period 的 volume °
- Residual gap : $\text{MAX}(\text{target renewable requirement} - \text{eligible existing procurement}, 0)$ °
- Over-coverage : 另行顯示，不以負 gap 混淆；需確認是否可 carry forward，不能自行假設 °

3. Existing procurement treatment

一份 contract 或 certificate record 至少區分：

Layer	Meaning	Decision use
Contracted	legal / commercial commitment	exposure and potential coverage
Expected	forecast delivery after known constraints	planning case
Delivered	metered / settled delivery	performance evidence
Eligible	accepted under approved policy / boundary	gap calculation
Claimed / retired	completed claim or use record	reporting / audit evidence

v0.1 model 使用 contracted MWh × delivery factor × eligibility factor 形成 planning eligible MWh。這是計算架構，不代表 AMAT policy；兩個 factor 必須有 evidence / owner。

Double-count controls

- Stable instrument ID + site / allocation + year °
- 同一 generation / certificate volume 不得同時出現在 multiple instruments / entities °
- 若 contract volume 與 certificate record 來自不同資料源，先 reconcile 再相加 °
- allocation / transfer / claim status 不明者列入 conditional，不默認為 eligible °

4. Data-quality logic

Minimum tests

Dimension	Test	Default disposition
Completeness	included sites / periods 是否齊全	quantify missing range ; material gap condition G3
Uniqueness	site-meter-period / instrument-year 是否重複	stop affected aggregation until resolved
Reconciliation	analysis total 是否 tie 到 owner-approved control total	document variance and approval
Validity	$\text{MWh} \geq 0$ 、target 0–100%、dates /	reject or correct invalid values

Dimension	Test	Default disposition
	units valid	
Lineage	source、version、transform、owner 可追溯	do not use untraceable data as approved evidence
Consistency	site / instrument mapping 與 boundary 一致	re-map or explicitly exclude

Quality state

- PASS：足以支持 intended decision。
- PASS WITH CONDITION：可用，但 output 必須帶 range / caveat / owner action。
- FAIL：不可用於 intended decision。
- NOT TESTED：尚未執行，不等於通過。

5. Core calculations

對每年 y ：

1. $Gross_Load_y = \Sigma \text{ approved site forecast / actual MWh}$
2. $Target_RE_MWh_y = Gross_Load_y \times Target_y$
3. $Eligible_Existing_y = \Sigma Contracted_MWh_{i,y} \times Delivery_Factor_{i,y} \times Eligibility_Factor_{i,y}$
4. $Residual_Gap_y = MAX(Target_RE_MWh_y - Eligible_Existing_y, 0)$
5. $Coverage_y = MIN(Eligible_Existing_y / Target_RE_MWh_y, 100\%)$
6. $Over_Coverage_y = MAX(Eligible_Existing_y - Target_RE_MWh_y, 0)$

若 denominator 或 target 未核准，結果標為 TBD / DRAFT，即使 formula 可運算。

6. Sensitivity design

v0.1 至少測試四個 cases：

Case	Load factor	Existing delivery factor	Purpose
Base	100%	100%	approved / current planning inputs
Higher load	105%	100%	demand / forecast upside
Lower delivery	100%	90%	project / certificate under-delivery
Combined stress	105%	90%	gap range and procurement buffer

正式 range 不應只使用固定 5% / 10%；應由 forecast uncertainty、historical variance、contract evidence 與 client risk appetite 更新。Demo factors 不是 AMAT assumption。

7. Management summary design

Management Summary 只呈現：

- approved / draft boundary 與 truth status ;
- first target year 、base residual gap 、combined-stress gap ;
- largest coverage expiry / step change ;
- material data-quality conditions ;
- 三個需要 client decision 的問題 ;
- handoff to 1.5 的 annual gap table / range 。

不在 summary 放所有資料欄位或假精確的小數。詳細 lineage 、source 與 mapping 留在 model / data tracker 。

Illustrative decision brief — SYNTHETIC

> Demo model 在 2028 年計算出 target requirement 105,000 MWh 、planning eligible existing procurement 52,250 MWh 。因此 base residual gap 為 52,750 MWh ; combined stress case 較高。這只證明 model 可跑，不代表 AMAT 的用電、target 或採購缺口。

建議的 G3 decision wording：

- Approve / approve with conditions / reject calculation boundary 。
- Approve named target curve and policy source 。
- Approve existing procurement treatment and factor owners 。
- Authorize annual gap / range as 1.5 demand input 。
- Assign unresolved evidence with owner / latest-needed date 。

8. Review protocol

Analyst first draft

先提交：boundary card 、兩個年份、input lineage 、base + one stress case 、三個最大的 unknowns 。不要等完整 site history 才第一次 review 。

Project lead review

以 client lens 問：

- 如果數字錯，哪個 decision 會受影響？
- 哪個 assumption 被藏在 formula ？
- output 是否區分 contract / delivery / eligibility / claim ？
- 1.5 是否能直接使用 annual gap / range ？

Client review

由各 owner 分別核准 data 、policy 、commercial treatment ; project sponsor / decision body 核准 G3 。不把沉默視為 approval 。

9. Handoff to 1.5

1.5 應收到：

- annual base gap 、stress gap 、target / load curve ；
- site / technology / timing constraints (若有核准資料) ；
- existing procurement expiry / uncertainty ；
- data-quality condition 與不能使用的欄位 ；
- approved criteria owner 與 date 。

1.5 不應重新發明 gap boundary ；若改變，必須以 assumption / change log 返回 1.4 並重新核准。

10. What this methodology does not conclude

- 不證明任何 AMAT site / load / target / policy fact 。
- 不判定任何 procurement instrument 一定符合 AMAT 或 RE100 policy 。
- 不提供法律意見、會計意見或 assurance 。
- 不以 synthetic sensitivity 代替正式 forecast / contract evidence 。